Tioluwani Akinloye

Dual Citizenship: Canada and Nigeria | **Native English Speaker**

587-566-2555 | akinloyetioluwani@gmail.com | LinkedIn | GitHub | tioluwaniportfolio.netlify.app

Education

Simon Fraser University - Master of Visual Computing, Expected 2028' Burnaby, BC
- *Relevant Coursework:* Distributed Cloud Systems, Machine Learning

University of Alberta - BSc Honors Computing Science, 2025' Edmonton, AB
- *Relevant Coursework:* Program Analysis, Models of Computation, Software Engineering, Data Structure & Algorithms

Experience

Simon Fraser University - Research Assistant Sep 2026 - Present
- Designing and building software architecture for joint-link simulation models as part of Project 15004, an experimental study on virtual model development.
- Applying open-source software design and development practices, including structured tutorials and documentation workflows, to build maintainable simulation tooling.
- Designing and evaluating software-driven interaction methods for testing and validating virtual models.

ScaleupMap - Full Stack Developer Feb 2025 - May 2026
- Built full-stack features using React, TypeScript, Go, and PostgreSQL for a platform capable of serving up to 1,000+ users.
- Engineered multiple Python-based external AI services leveraging Google Gemini, with APIs deployed on GCP.
- Designed and optimized 20+ relational database schemas using DBeaver, supporting scalable data workflows.
- Implemented multiple unit, API, and E2E tests; through regular code review, strengthened defensive coding and modular design practices, and reduced redundant API calls by up to 25%.

Knead Technologies - Student Developer Jan 2024 - April 2024
- Designed an intuitive dashboard and implemented various features, including on-demand and routine donation tracking, using ReactJS (Typescript), PHP (Laravel), Apache, and MySQL, all running on Docker, capable of serving 500+ active users.
- Developed the closed marketplace page and designed UI/UX components to streamline user transactions and feature key metrics.
- Conducted E2E testing of the application & APIs using Cypress, & unit testing with PHPUnit, reducing production bugs by $\sim$20%

Projects

Networking App Startup - Co-Founder & Technical Lead
- Co-founded and led technical development of a mobile app startup, leveraging LLMs and data contextualization to help users build their professional network
- Owned the product pipeline end-to-end: validating the concept, designing UI/UX through Figma prototypes, and defining architecture and UML class diagrams to guide the team
- Evaluated frameworks and languages to implement a custom stack (Supabase, Flutter/Dart, FastAPI, Gemini, GCP), integrating services through GCP with rigorous testing and data security

SmartVision - Full-Stack ML Training App
- Built a full-stack web application enabling users to train, evaluate, and interact with machine learning models in real time
- Designed a canvas-based input feature and backend model-serving pipeline supporting multiple model architectures with optimized training and inference times
- Built using Vue.js for the frontend, Flask for the backend API, and PyTorch for model implementation and training

StudyAI - Full-Stack Learning Platform
- Developed a full-stack learning platform that generates an AI chatbot tailored to the content of an uploaded video
- Built using React.js for the frontend, FastAPI for the backend, and Firebase Realtime Database for data storage, allowing users to upload videos or YouTube links and generate transcripts, summaries, and related content
- Integrated OpenAI's Whisper model for transcription and OpenAI's API to power a chatbot that answers questions, summarizes content, and generates quizzes

Skills

Languages | Python, Java/TypeScript, Golang, C++, PHP, C, Racket, R
Technologies | Git, Docker, Postman, Cypress, OpenGL, OpenCV, PyTorch
Frameworks | React.js, Node.js, Express.js, FastAPI, Flask, Laravel, Vue.js
Concepts | OOP, TDD, REST, Agile, Scrum, CI/CD, UI/UX, MERN, MEVN

Additional Information

Awards: Academic Honors (2020), Distinguished Service Award (2020)
Certifications: International Baccalaureate Diploma (2020)
Achievements: Faculty of Science Dean's List (2023)